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Intelligent forecasting and distribution in cross-border e-commerce import trade: A deep-learning-based iterative optimization approach

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ABSTRACT

The dramatic growth of cross-border e-commerce (CBEC) trade promotes the vigorous development of bonded warehouses, providing overseas suppliers with an opportunity to lay out distribution networks to meet the domestic consumers' growing logistics efficiency requirement. This paper considers the distribution network design problem with iteratively updated demand. Specifically, we first construct a hybrid deep learning model, which integrates a convolutional neural network for extracting recessive features and long short-term memory for a retrograde time extension to forecast the consumers' demand. Then, a mixed integer linear programming (MILP) model is developed to formulate the distribution network design, which aims to rent the appropriate storage capacities of some warehouses in different locations and make the product allocation plans with minimum operation cost. The Benders decomposition algorithm is appropriately adopted as the solution approach to the proposed model. When the warehouse locations and distribution plan are initially developed, the logistics timeliness of some destinations will be improved, potentially leading to a redistribution of consumers' demand. Therefore, we integrate the prediction and MILP models to construct a forecasting-distribution iterative optimization process to explore the optimal solution dynamically. A real case study is used to verify the effectiveness of the proposed integrated approach. Our research formulates characteristic distribution network design solution for overseas suppliers engaged in CBEC import trade, providing valuable insight to achieve an iterative optimization process through organically linking deep-learning-based forecasting and optimization processes.

1. Introduction

Driven by the acceleration of economic globalization and Internet technology, global cross-border e-commerce (CBEC) has witnessed tremendous growth in recent years [1], which is expected to increase to $4856 billion by 2027 [2]. The dramatically growing market demand has driven the rapid development of many CBEC platforms, such as Amazon, Tmall Global, JD Worldwide, eBay, etc., to satisfy the continuously increasing demand of Chinese online consumers for international commodities. However, the indispensable long distance headway transportation from overseas origin to domestic port makes imported commodities generally generate lower logistics timeliness and higher cost for fulfillment in traditional cross-border trade. Therefore, how to improve the logistics efficiency of CBEC import business to meet the demand of online consumers for logistics service quality has become one of the most important challenges.

For cross-border import logistics mode, the bonded warehouse and direct mail are two major logistics modes overseas suppliers can choose. The bonded warehouse mode allows retailers to integrate and transport large quantities of commodities to the import port and store them in the bonded warehouse of the comprehensive bonded zone, effectively saving logistics cost and guaranteeing product quality but potentially facing the inflexible inventory management problem in the form of stockout or excess [3,4]. For direct mail mode, overseas retailers reduce the stockout or excess inventory cost by shipping commodities directly from the place of origin to local customers but have to burden the high transoceanic logistics cost for each order [5]. When customers place orders online, the bonded warehouse only needs to send clearance applications to the Customs and Excise Department, and the domestic fulfillment can be delivered after clearance is completed. This one-stop logistics service, which eliminates the time-consuming process of overseas direct mail and integrates warehousing and domestic drop shipment, has enabled the bonded warehouse logistics mode to be rapidly and widely applied in CBEC import business.

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Meanwhile, the logistics timeliness is one of the significant factors affecting online consumers' purchasing behavior [6,7], because online consumers are more sensitive to the logistics service quality than brick-and-mortar channels [8]. Since the cross-border imports must be stored in comprehensive bonded zones, imported commodities cannot be deployed to almost every city, county, or even community like domestically produced goods. However, the widespread deployment of bonded warehouses has laid the foundation for the inventory distribution of imported goods in recent years. For instance, Cainiao Network, a leading 4PL service provider in China, has launched the nationwide import bonded goods distribution service since 2019 to further improve the logistics fulfillment efficiency.[1] The increasing demand of consumers for logistics timeliness has prompted many CBEC retailers to spare no effort in seeking any way to deploy their products to bonded warehouses closer to consumers, which would increase sales volume as much as possible by further improving logistics service quality [9].

Generally, the platform self-operated and the merchant-operated mode are the two main business modes on the CBEC platform. The former refers to the e-commerce platform operating an online store, under which the overseas retailers are converted into suppliers only responsible for wholesaling commodities to the platform, and the e-commerce platform is responsible for the operation and sales rights of the commodities. The merchant-operated mode represents the overseas retailers opening their own stores in the e-commerce platform. Under this mode, the e-commerce platform only provides online traffic and platform services, and the retailers need to manage inventory and operations while making sales plans by themselves. Even though the self-operated mode of the CBEC platform can provide centralized distribution service to decline the logistics timeliness, the unified distribution of multiple products from several merchants results in lower profit for a single merchant while also incurring high entry costs to enter the self-operated channel. Thanks to the one-stop warehousing and drop shipment services provided by bonded warehouses, the retailers do not have to invest to build their own warehouses. They can simply rent their required inventory capacity from the target warehouse on a monthly basis, which greatly stimulates many CBEC retailers to design their own personalized distribution strategy. On the one hand, distribution strategy brings dynamism and timeliness, but may also results in higher cost, such as additional warehouse rental cost and product transshipment cost. On the other hand, the selling price of imported commodities is tax-inclusive (including tariff, consumption tax, etc.), so the prices are generally higher than domestic commodities [10]. Thus, the sales volume of imported commodities in different provinces are closely associated with the local residents' income level. Moreover, the economically developed provinces have complex logistics networks, large freight flows, and strong delivery capabilities. Even for imported commodities with relatively low logistics efficiency, the logistics service providers in developed provinces can basically achieve the ability to fulfill orders of one-day shipping within the province and two-day shipping in adjacent provinces but may achieve little effect for the distant central and western provinces. Hence, it is a challenge for overseas retailers to make a reasonable trade-off between the potential sales stimulation and the additional cost increase brought about by the implementation of characteristic inventory distribution strategy.

Motivated by the aforementioned challenges, we construct a hybrid deep-learning-based optimization model to forecast the potential demand and design the distribution scheme for the sales season through an iterative optimization procedure, and make the following contributions:

- We extract five data dimensions of the historical transaction data as the input data set to train the demand predictor. Many studies on demand forecasting in online shopping tend to ignore the role of online comments or fail to quantify the utility of comments. In the current work, we propose an online comment score modification approach that integrates sentimental assessment through the natural language processing technique.
- We propose a comprehensive deep learning approach, that integrate convolutional neural network (CNN), long short-term memory (LSTM), and genetic algorithm (GA), to train the time-series historical transaction data and generate the demand forecasting results. In simple terms, CNN aims to extract recessive features, LSTM is used to retrograde time extension and generate the prediction results, and GA is adopted to optimize the hyper-parameters of CNN and LSTM.
- We develop a mixed integer linear programming model to implement the inventory distribution strategy, and adopt the Benders decomposition algorithm to solve the optimal plan. The proposed model can well serve the practical requirement of overseas retailers for renting warehouses in different location and designing product inventory distribution scheme.
- Considering the consumers' demand is sensitive to logistics efficiency in CBEC import business, the implementation of the distribution plan may significantly improve the logistics response time of some consumers closer to the newly rented warehouses, which may stimulate the increase of potential demand. We further build a forecasting-distribution iterative optimization procedure to formulate the demand reforecasting and distribution network redesign process until the distribution network structure no longer changes.

The rest of this paper is organized as follows. We review the relevant literature in Section 2 and present the detailed problem description in Section 3. Section 4 presents the proposed deep learning approach for demand forecasting. Section 5 shows the mixed integer linear programming model, solution approach, and the iterative optimization procedure. Section 6 is allocated to a real-world case study to verify the proposed model. The final section is the conclusion.

2. Literature review

2.1. Cross-border e-commerce supply chain

The rapid growth of cross-border e-commerce trade has promoted some researchers to work on the operation management problems of the CBEC supply chain, mainly focusing on tariffs and the choice of overseas logistics modes. For instance, Niu et al. [10] explored the influence of tariffs on two retailers engaged in imported business when selling two different products with quality-differentiated and substitutable attributes. Shi et al. [1] studied the operation decision of an international brand to expand new overseas markets with considering of multinational tariff rules while considering multinational tariff rules. Moreover, the choice of CBEC logistics modes has also been further concerned in recent years. Wang et al. [11] proposed three cross-border logistics models for overseas retailers to select the optimal cross-border logistics mode considering return strategy. Further, some researchers have investigated the selection of logistics modes under different external environments. Niu et al. [2] investigated the strategic waiting choice for overseas retailers operating two channels of bonded warehouses and direct mail under the disruption risk to manage the inventory more flexible. Zha et al. [3] modeled a CBEC supply chain to investigate the information-sharing strategy of CBEC platforms on overseas merchants' logistics mode choice of bonded warehouse or direct mail. Zhang et al. [12] studied the logistics mode selection of overseas retailers between bonded warehouse and direct mail, which has different logistics costs and inventory risks, and also explored the information sharing strategy of CBEC platform on retailers' logistics mode choice.

Some researchers have studied many classical operation management issues under the CBEC business scenario. Han et al. [13] studied the impact of free shipping strategy on consumers' purchasing behavior and profit of CBEC platforms to develop an appropriate free shipping strategy for the CBEC platform based on the real operating data. Niu

et al. [4] explored the effect of blockchain technology on quality verification in CBEC platforms, indicating that blockchain might increase the wholesaling profit but decrease the retailing profit of overseas companies. Jiang et al. [14] considered the booming development of gray markets such as overseas shopping on CBEC platforms and studied the impact of gray markets on global supply chain management through developing an analytical model. Additionally, some scholars have investigated many different operation management problems from various perspectives, such as disruption analytics [15], inventory risk management [16], real-time payment in CBEC transactions [17], CBEC platform data set fulfillment [18], customs declaration [19], etc., to propose valuable management insights into CBEC platforms and overseas retailers.

2.2. Intelligent forecasting

With the dramatic development of artificial intelligent techniques, many advanced forecasting methods and tools have been widely adopted to address some practical prediction problems in many fields. For example, Mamoudan et al. [20] applied machine learning and game theory to forecast the price of perishable products dynamically. Nenova and Shang [21] constructed an intelligent forecasting model to carry out accurate fast disease trajectory predictions based on electronic health records. Kraus and Feuerriegel [22] built a structured-effect neural network to estimate the remaining useful life of machinery, infrastructure, or other equipment. With the work on demand prediction becoming the hotspot recently, Crivellari et al. [23] constructed a multi-target deep learning model to predict the consumer demand for short-term fresh food. Wang et al. [24] built an integrated forecasting model to predict the power demand in a multi-heterogeneous data environment. Van Nguyen et al. [25] developed an advanced data-mining prediction model to forecast the consumer demand for remanufactured products, and the proposed approach addressed the complex and non-linear difficulty in demand prediction.

The demand forecasting for online sales, on the other hand, exhibits many characteristics that differ from offline brick-and-mortar channels, e.g., special events and promotions, online reviews, delivery time, and convenient price comparisons [25–27]. Omar et al. [28] proposed a new approach to forecast the demand in omnichannel, driven by data on customers shopping baskets. Kharfan et al. [29] adopted machine learning techniques to help online fashion retailers improve the prediction accuracy of future demand by fully taking characteristics of varying preferences, short lifecycles, and popularity. Ren et al. [30] developed a deep-learning-based decision-making model that integrated inventory optimization and demand prediction for third-party forwarding logistics providers to optimize the random arrivals of orders on CBEC platforms. Considering that the prediction at the aggregation level could lead to real-time information loss, Dai et al. [31] proposed an adaptive hierarchical incremental forecasting (AHIF) approach, comprising an incremental approach and a hierarchical process to handle structural variation and extract relevant real-time information to predict the daily demand of retailers engaged in O2O business.

2.3. Distribution network optimization

The core process for a distribution network design lies in the hub location and inventory allocation [32]. Lin and Lee [33] studied the hub network configuration problem for time-definite less-than-truckload carriers and formulated a mixed integer programming solved by implicit enumeration with an embedded pricing subproblem. Dukkanci and Kara [34] developed a hierarchical multimodal hub network structure that consists of three layers and optimized the allocation flow of the distribution network by designing a heuristic solution algorithm based on the subgradient approach. Shahabi and Unnikrishnan [35] developed a robust mixed integer nonlinear programming model to optimize the hub location under uncertain demand scenarios. Taherkhani and Alumur [36] investigated the profit-maximizing hub location problems by considering three characteristic allocation plans: single, multiple, and r-allocation. Azizi and Hu [37] extended the classical location-routing problem by integrating direct shipment mode to optimize the distribution network for multi-product pickup and delivery. In terms of data-driven demand forecasting and distribution network optimization, Liu et al. [9] developed an iterative prediction-and-optimization for distribution network design in E-commerce trade. Qin et al. [38] proposed a two-stage stochastic programming method for online retailers to design their distribution network that integrated data-driven demand predictions to realize the optimal benefit between reduced delivery time and increased inventory cost. Finally, although many literatures have studied a series of network design problems, few of them have investigated the dynamically updated network design problem that integrates the attribute of online demand is sensitive to logistics timeliness, especially in the CBEC import business.

Since hub location and inventory allocation can be decomposed into the typical master problem and subproblem, i.e., the subproblem (inventory allocation) is optimized based on the master problem (hub location), the Benders decomposition (BD) algorithm has been a classical approach for researchers to address the distribution network design problems in various scenarios. For instance, Oliveira et al. [39] presented a novel formulation of multiple allocation hub network design problems and adopted the BD algorithm to find the optimal strategy. Zheng et al. [40] constructed a supply chain network and proposed a generalized BD algorithm to optimize the location, inventory, and routing problem integrally. Additionally, the classical hub location and inventory allocation problem can be extended to other more complex scenarios, such as closed-loop supply chain network design, multi-echelon multi-product distribution network design, multi-period capacitated location-routing network design, multi-tier network system design, etc., where many different forms of the BD algorithm have been proven to be effective [41–44].

2.4. Research gap

Through a brief review of the relevant literature, we have identified the following research gaps, which will be addressed in this work.

- Numerous studies on cross-border supply chains have focused on logistics modes, blockchain technology, disruption risk, and data set fulfillment, highlighting a lack of specialized research on CBEC import trade distribution networks.
- Many studies fail to fully integrate the characteristics of CBEC import for a more effective demand forecasting process, and the application of integrated deep learning models is still insufficient.
- Previous research on location and routing optimization problems of distribution networks has mainly been implemented based on simulation data or static cases, failing to organically integrate distribution network design into demand forecasting procedure to perform a dynamic iterative optimization process.

3. Problem description

This section provides a detailed introduction to the research problem and the key preliminaries throughout the paper. The detailed notations and model specific to the deep learning framework and distribution optimization model will be introduced in later sections.

3.1. Distribution network

This work investigates the optimization of imported product network distribution for CBEC retailers to improve logistics service quality.

We design a distribution network for overseas retailers to simulate the actual operation scenario CBEC imported commodities and verify the effectiveness of the proposed model and approach through a real-life case. Since the CBEC imported commodities must be stored in the bonded warehouse in the comprehensive bonded zone [2], the research objective evolves into a location selection and product distribution optimization problem within a given optional collection. According to the geographical advantages and the development of cross-border e-commerce for different regions, we have selected some locations as the transshipment and distribution warehouses (TDWs) to collect commodities from suppliers whose locations are given, deliver commodities to the consumers in near cities, undertake the mass transshipment between TDWs, and distribute commodities for the delivery-only warehouses (DOWs) to balance the inventory allocation. We consider an overseas retailer forecasts the demand of each consumer location within a specific sales period in the future based on historical sales volume and a series of collated data and then makes the distribution plan. Then, the retailer needs to select the optimal locations of TDWs and transport the total predicted demand to them. After retaining the inventory of some consumers' predicted demand that should be delivered by the TDWs, the remaining inventory will be allocated and transferred from TDWs to DOWs to satisfy other consumers' predicted demand in cities around DOWs. We assume that the retailer is allowed to conduct horizontal transshipments between TDWs, while the transshipments between DOWs are prohibited.

3.2. Iterative optimization

Since logistics timeliness is one of the main factors significantly affecting consumers' purchase intention [6], the retailer has to distribute and transport products to some e-commerce bonded warehouses closer to consumers' points before the sales period. The location of warehouses for the logistics timeliness enhancement may also affect consumer demand [45]. Hence, we define an interrelated interdependence model that integrates iterative optimization on demand prediction and distribution network realignment. First, the historical sales data, logistics service quality data, time-series comments that modify the original score by integrating sentiment analysis, and statistical data measuring purchasing power at the city level are collected to construct a hybrid CNN-LSTM model and forecast the consumer demand for a certain period in the future. Second, given a set of suppliers, consumer demand distribution, and TDW and DOW candidate locations, the location-routing optimization problem is built to select the optimal set of TDWs and DOWs and distribution volume of commodities that would minimize the total cost. Notably, after implementing the distribution strategy, some inventory will be allocated to several leased bonded warehouses, so the delivery time for consumers in closer regions to these warehouses will be greatly improved. Further, the shorten delivery time for some consumers may stimulate the increase in the potential demand. Hence, the updated delivery time needs to be re-input into the demand prediction model to generate the newest demand distribution result, and then, a new round of location optimization will be implemented based on the newly obtained demand distribution. In general, the core of forecasting-distribution iterative optimization (FDIO) strategy lies in the fact that if the logistics timeliness is changed due to the distribution network establishment, all consumers' demand distribution should be re-forecasted until the entire distribution network no longer change. The research framework is illustrated in Fig. 1.

4. Deep learning approach

In this section, a multi-source information fusion-driven CBEC imported product demand prediction model is constructed to train the multi-source heterogeneous time series data, especially the text data, and generate demand prediction results. In the proposed hybrid deep learning approach, we use a convolutional neural network (CNN) to extract the recessive features of time series data. Then, we input the preprocessed data into the prediction module of the long short-term memory (LSTM), which shows the superior training performance on time sequence data. Finally, the hyper-parameters of the hybrid CNN-LSTM prediction model are further optimized using a heuristic algorithm to reduce errors as much as possible. For the collected product data, including price and sales volume, we preprocess sales transactions data from the dimensions of chronological sequence and delivery address. Then, we extract the city-level data for each delivery address to measure the purchasing power, such as per capita disposable income, population, urbanization rate, etc. We also collected the logistics service data, including the average delivery time for each consumer, which is the core to realizing the iterative optimization process. However, the operation data of the merchant cannot be ignored, such as the number of fans, damage rate, promotional activity, etc., requiring collection and collation in chronological sequence. Additionally, the online comment is also one of the critical factors significantly affecting consumers' purchase intention, so we design a comment score modification procedure to extract the online consumers' preference and regard the online comment as one of the key attributes that integrates to the training set.

4.1. Comment score modification by sentimental assessment

The implicit features of customers' feedback in product comments have been demonstrated to be an undeniable feature that may affect the subjective preference of potential customers significantly [46]. The effect of product comments on the purchasing choice of potential consumers is not only shown in the product ratings but also in the product characteristics and sentiments described by the consumers. However, many e-commerce platforms, e.g., JD and Suning, only allow the selection of integers between 1 to 5 in their online comment rating systems, making it difficult to assess consumers' preferences through purely integer ratings. One of the notable contributions of this research proposing a method for the original comment score modification to incorporate sentiment assessment.

To extract the key product features from the comment text, the first step is to identify the key statements of purchased consumers' perception of product features. The product feature words in reviews are generally verbs, nouns, and noun phrases, while opinion words are typically adjectives, adverbs, verbs, and nouns. For example, the common combination of "noun+adjective" includes "good quality", "fast logistics", "complete packaging", etc. Thus, we design a feature-opinion word pairs extraction algorithm that generalizes to part of speech granularity and discovers the connections based on statistical methods, thereby improving the recall rate of feature and opinion word extraction. Due to the strong domain correlation of feature words for a specific product, it is necessary to conduct the word frequency statistics of the feature words and the clustering of synonyms after extracting the feature words. By evaluating the accuracy and recall rates, more comprehensive feature-opinion word pairs are extracted to preserve the key product features. Furthermore, the more times a word appears in a comment, the greater the importance of that word to the comment, reflecting the higher perception of the consumer who posted the comment. In a document set, the frequency of different product feature words often indicates the different levels of consumers' attention, also highlighting the various preferences of the product features. Then, we designed an algorithm based on term frequency–inverse document frequency (TF-IDF) to assess the consumer preferences for various features, generating weight vectors w_i for each feature word. Considering the paper's length, we only present the brief steps for illustration in Algorithm 1 while providing more important core steps for the original comment score modification in the main text, which integrates sentiment analysis. The detailed steps and algorithm pseudocode in this sub-section are available from corresponding author.

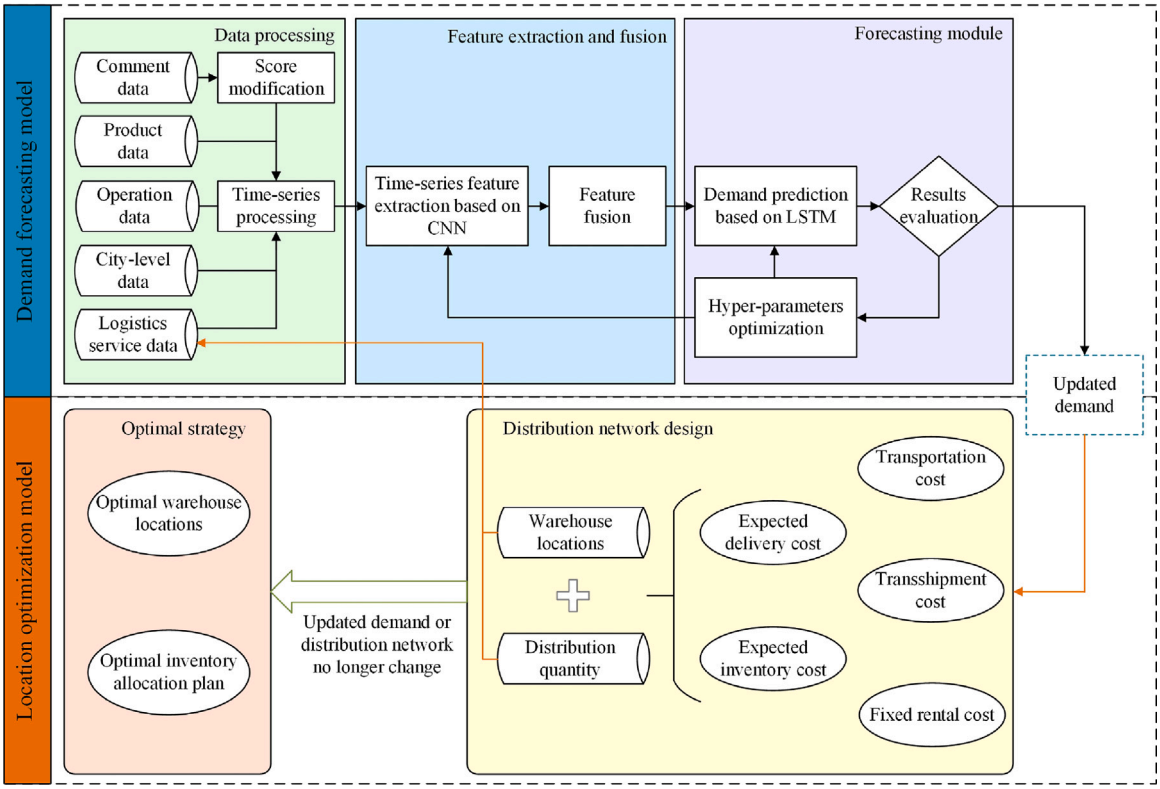


Fig. 1. The research framework.

Algorithm 1: Brief steps for feature extraction and preference estimation

Data: comment set $\mathcal{C}$;
Result: feature preference vector set $\mathcal{V}$, feature weight set $\mathcal{R}$, opinion word set $\mathcal{Q}$.

1 Extract the word and part of speech in comment S_k as part of speech template.
2 Delete all words in the template but keep their lexical labeling.
3 If the length of the candidate template is less than the length threshold β and it matches a certain template N_x in the template set, so the frequency of the candidate template N_x will plus one; When the candidate template is different from the candidate template, its frequency is set to 1 and added to the template set.
4 Calculate the frequency of each part of speech template.
5 For all candidate templates in the template set, if their frequency p_{N_x} is less than the frequency threshold γ, then delete the part of speech template from the template set.
6 Match the part of speech template in each comment with the template set, and extract the corresponding feature-opinion word pairs.
7 The filter and cluster of feature words, domain feature words, and corresponding synonyms.
8 Calculate the TF-IDF values of all feature words in the comment set.
9 Calculate the feature preference as the weight vector w_i.

We referred to the sentiment dictionary (i.e., Boson social media text emotional thesaurus) to calculate the sentiment value of the feature dimension of the comments. For a comment sentence S_k, each sentence included one or more feature word $f_i \in \mathcal{V}$. The emotional intensity of S_k can be obtained by calculating the emotional intensity of all feature words. We used $d(o_j)$ to represent the emotional degree of the opinion word $o_j \in \mathcal{Q}$, with a value range of [−5,5]. The positive and negative values of emotional degree represent the positive and negative meanings of the feature word, respectively. If o_j does not exist in the sentiment dictionary, the value is 0. Let $w(o_j)$ represent the weight of the degree adverb that modifies the opinion word o_j, with an interval of (0,2). Here, $u(o_j) > 1$ indicates emotional enhancement, while $u(o_j) < 1$ refers to emotional weakening. If there is no corresponding degree adverb, then $u(o_j) = 1$. Let $n(o_j)$ represents the negative adverb reversal coefficient. When there is an odd number of negative adverbs, $n(o_j)$ takes a value of −1, indicating the reversal of emotional tendency. When there is an even number of negative adverbs or no negative adverbs, then $n(o_j) = 1$. Considering that many opinion words with low emotional degree will cause great interference to the comment, a discrimination threshold value 0.2 can also be set for a single sentiment word, and only the significant opinion words can be reserved, as shown in Eq. (1):

$$v(o_j) = \begin{cases} 0, o_j \notin \text{Sentiment dictionary} \\ 0, \left|d(o_j)\right| \le 0.2 \\ d(o_j), \left|d(o_j)\right| > 0.2 \end{cases} \tag{1}$$

The emotional intensity of feature f_j can be calculated through Eq. (2):

$$e(f_j) = n(o_j) \times u(o_j) \times v(o_j) \tag{2}$$

Further, the emotional intensity E_k of comment S_k can be obtained, as shown in Eq. (3):

$$E_k = \sum w_i \times e(f_i) \tag{3}$$

Then, we should modify the comment score that integrates emotional intensity to a specific range of intervals. We can obtain the emotional intensity of all comments $E_k \in [E_{\min}, E_{\max}]$ through previous steps. Given a correction range $[M_{\min}, M_{\max}]$, i.e., $[M_{\min}, M_{\max}] =$

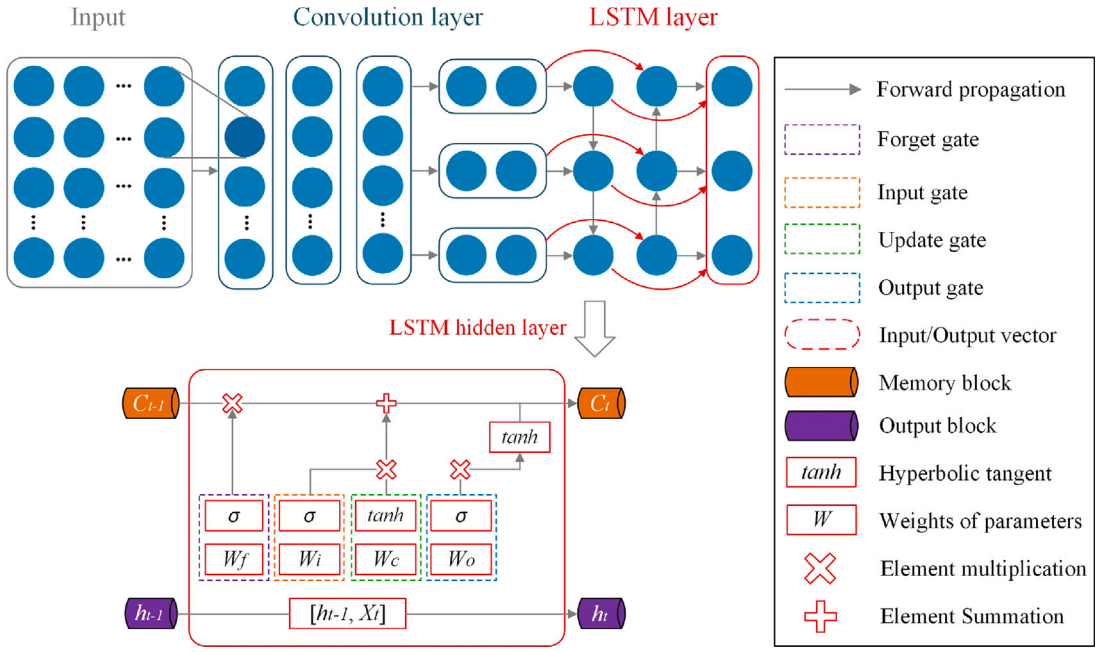


Fig. 2. The schematic diagram of CNN-LSTM.

$[-\tau, +\tau]$ and $\tau = 0.5$, any numerical value of E_k on the original interval $[E_{\min}, E_{\max}]$ can be transferred to the target range by using the standardized formula:

$$M_k = \frac{M_{\max} - M_{\min}}{E_{\max} - E_{\min}} \times (E_k - E_{\min}) + M_{\min} \tag{4}$$

The modified comment score r'_k can be obtained, shown as below.

$$r'_k = r_k + M_k \tag{5}$$

where r_k and M_k represent the original score of comment k, and the modified value, respectively. Further, by classifying comments in chronological order and averaging the modified scores for the same posting time, we can obtain the comprehensive influence of the comment utility, which can be the input source for the following prediction model together with other collated data.

4.2. CNN-LSTM for demand forecasting

The convolutional neural network is an extraordinary neural network that shows a strong capacity to process data in the form of multiple arrays [47]. Long short-term memory networks, proposed by Hochreiter and Schmidhuber [48], are widely used to deal with some specific drawbacks of Recurrent Neural Networks (RNN), which can be seen as an instance of the broader class of RNN [49]. In this section, we fully integrate the capability of convolutional layers of CNN on extracting recessive sequential characteristics and the advantage of LSTM layers in exploring short- and long-term relevance [50,51]. Specifically, the core idea of this hybrid deep learning approach is to use the CNN layer to reshape the raw input data into a more convenient representation format, attenuating the noise in multidimensional sequence, and to employ LSTM layers to efficiently capture sequential pattern information [23], as illustrated in Fig. 2.

CNN layer. CNN includes a series layer of input, convolutional, pooling, fully connected, and output. The preprocessed data are transformed in the input layer. Then, the time-series features of the input data set are extracted using convolutional kernels of different sizes. Further, the extracted features are integrated into the pooling layer to form the implicit feature set and generate the output after receiving the properties. The most characteristic and significant step of CNN is focused on extracting recessive features from the input data set in the convolutional layer. The number of convolutional kernels K defines the $K-$dimensional output or sequence equivalently. The convergence operation is used in the convolutional layer to extract the feature map instead of the general matrix operation [20]. Eqs. (6) and (7) are used for the convolutional layer.

$$\mathbb{O}_j = \sum_i \mathbb{X}_i \times \mathbb{K}_{ij} + \mathbb{B}_j \tag{6}$$

$$\mathbb{F}_j = a(\mathbb{O}_j) \tag{7}$$

where $\mathbb{O}_j$ represents the output of the convolutional operation, $\mathbb{X}_i$ is the input data matrix, $\mathbb{K}_{ij}$ refers to the convolutional kernel, $\mathbb{B}_j$ denotes the bias, $\mathbb{F}_j$ indicates the feature map, and $a(\bullet)$ is the activation function.

Then, the pooling layer follows to reduce the feature map resolution, enhancing the efficiency and reducing the computational complexity. The pooling layer filters out the maximum value from each neuron cluster to address the over-fitting problem [30], shown as below.

$$\mathbb{Z}_j = \max(\mathbb{O}_j) \tag{8}$$

LSTM layer. LSTM shows a strong capacity to address the vanishing gradient problem, as the network can still bridge long-term information effectively even with the increase of layers and iterations [52]. The main structure of the LSTM consists of a state vector and three gates, including input, forget, and output, which are used to realize the temporal features by updating or eliminating historical information.

The forget gate f determines which past information to eliminate from the state vector. It checks the previous output h_{t-1} and current input X_t and decides what information of the state vector C_{t-1} should be discarded, shown as follows:

$$f_t = \sigma(W_f \bullet [h_{t-1}, X_t] + b_f) \tag{9}$$

where σ, W_f and b_f represent the activation function of *sigmoid*, the weight matrix, and bias term, respectively. After eliminating part of the previous information, the neural network needs to supplement the newest information to the state vector, which is the role of the input gate. The first part is a *sigmoid* layer that determines which information should be updated, and the second part is a *tanh* layer that establishes a new candidate vector for $\tilde{C}_t$. Then, these two parts update the state vector jointly.

$$i_t = \sigma(W_i \bullet [h_{t-1}, X_t] + b_i) \tag{10}$$

$$\tilde{C}_t = \tanh\left(W_C \bullet \left[h_{t-1}, X_t\right] + b_C\right) \tag{11}$$

$$C_t = f_t \times C_{t-1} + i_t \times \tilde{C}_t \tag{12}$$

where i_t and $\tilde{C}_t$ are two kinds of information updated by the input gate, W_i and W_C are weight matrix, b_i and b_C are deviation items, C_t represents the updated state vector. Finally, the output gate is used to selectively control the output, which can be obtained by the O_t and *tanh* activation function multiplication.

$$o_t = \sigma\left(W_o \bullet \left[h_{t-1}, X_t\right] + b_o\right) \tag{13}$$

$$h_t = o_t \times \tanh\left(C_t\right) \tag{14}$$

4.3. Hyper-parameters optimization

Although many previous works have verified the effectiveness of the CNN-LSTM algorithm on time-series forecasting [30,53], it is difficult and laborious to obtain the optimal set points of hyper-parameters, such as the number of kernels of CNN and neurons of LSTM if a specific optimization process is not implemented. Generally, heuristic algorithms can be chosen to optimize the set points of hyper-parameters; thus, we adopt the most commonly used genetic algorithm to get the optimal values. The individual fitness function is shown as follows:

$$\text{fitness} = \frac{1}{\text{MAPE}}\left[1 + w_{\text{CNN}}\left(1 - \frac{n_{\text{CNN}}}{n_{\text{Total}}}\right) + w_{\text{LSTM}}\left(1 - \frac{n_{\text{LSTM}}}{n_{\text{Total}}}\right)\right] \tag{15}$$

where MAPE is the mean absolute percentage error and n_{CNN}, n_{LSTM}, and n_{Total} represent the number of kernels of CNN, the number of neurons of LSTM, and the total number of kernels and neurons, respectively. Additionally, w_{CNN} and w_{LSTM} indicate the influential weight of the CNN and LSTM layers on hybrid network performance, set as $w_{\text{CNN}} = 0.75$ and $w_{\text{LSTM}} = 0.25$ in the present work.

5. Import network optimization model

This section presents the notations, formulates a mixed integer linear programming model to optimize the import network, and introduces the detailed solution approach.

5.1. Distribution network optimization

The notation and description used for the distribution network optimization model are summarized in Table 1. The basic assumptions of the MILP model are shown as follows:

1. Horizontal transshipment of commodities between TDWs is allowed, while DOWs are responsible for storage and delivery functions. Horizontal transshipment between DOWs is prohibited.
2. Since the TDW has to perform both horizontal transshipment and delivery functions, we assume a warehouse has to burden the rental cost and the transshipment cost if it is operated as a TDW. Moreover, the rental cost for renting warehouse is defined as the total sales period (converting days to monthly form) multiplied by the corresponding monthly warehouse rental cost.
3. The picking and package cost in the transportation process from suppliers to TDWs, the transshipment process between TDWs, and the delivery process of each warehouse are integrated into transportation or delivery costs, respectively.
4. We assume the expected length of the sales period is T, the expected daily sales volume of each consumer location is evenly distributed, and all of the distributed inventories will be sold out at the end of the sales period. Thus, the average daily inventory of warehouse $j \in \mathcal{W}$ is $\frac{\sum_{k\in\mathcal{K}} z^{jk}}{2}$ and the corresponding expected inventory cost can be written as $\frac{\sum_{k\in\mathcal{K}} z^{jk}}{2}\varepsilon^j T$.

Table 1
Notations of the optimization model.

Set	
$\mathcal{W}$	Candidate warehouse set $\mathcal{W} = \{1, 2, \ldots, W\}$
$\mathcal{S}$	Supplier set $\mathcal{S} = \{1, 2, \ldots, S\}$
$\mathcal{K}$	Consumer set $\mathcal{K} = \{1, 2, \ldots, K\}$
Parameters	
T	Expected sales period
D^k	The forecasting demand of the consumer $k \in \mathcal{K}$ over the entire sales period
θ^{ij}	Distance between node $i \in \mathcal{W}$ and $j \in \mathcal{W}$
U^j	The inventory capacity of DOW $j \in \mathcal{W}$
δ_a, δ_b	Maximum selectable number of TDW and DOW
κ^i	Fixed cost for transshipment through warehouse $i \in \mathcal{W}$
λ^j	Fixed cost for renting warehouse $j \in \mathcal{W}$
ε^j	Unit inventory cost at DOW $j \in \mathcal{W}$
μ^s, μ^i	Per kilometer transportation cost of transshipment started from supplier $s \in \mathcal{S}$ or TDW $i \in \mathcal{W}$
η^i	Per kilometer transportation cost of distribution started from TDW $i \in \mathcal{W}$ to DOWs
c^{jk}	Unit delivery cost from DOW $j \in \mathcal{W}$ to consumer $k \in \mathcal{K}$
Decision variables	
x^{si}	Number of goods transited from supplier $s \in \mathcal{S}$ to TDW $i \in \mathcal{W}$
y^{ij}	Number of goods distributed from TDW $i \in \mathcal{W}$ to DOW $j \in \mathcal{W}$
z^{jk}	Number of goods delivered from DOW $j \in \mathcal{W}$ to consumer $k \in \mathcal{K}$
$q^{ii'}$	Number of goods transshipped from TDW $i \in \mathcal{W}$ to another TDW $i' \in \mathcal{W}\backslash\{i\}$
u^i	Binary variable; 1 if warehouse $i \in \mathcal{W}$ is served as TDW
r^j	Binary variable; 1 if warehouse $j \in \mathcal{W}$ is served as DOW

5. The forecasting demands of all consumers have to be strictly satisfied.

The objective of the import product allocation network is to minimize the total logistics and operation costs. We formulate the objective function (16) and the constraints (17)–(28) as follows.

$$\min \sum_{s\in\mathcal{S}}\sum_{i\in\mathcal{W}}\left(\mu^s\theta^{si}x^{si}\right) + \sum_{i\in\mathcal{W}}\sum_{i'\in\mathcal{W}\backslash\{i\}}\left(\mu^i\theta^{ii'}q^{ii'}\right) + \sum_{i\in\mathcal{W}}\sum_{j\in\mathcal{W}}\left(\eta^i\theta^{ij}y^{ij}\right) + \sum_{j\in\mathcal{W}}\sum_{k\in\mathcal{K}}\left(c^{jk}z^{jk}\right) + \sum_{j\in\mathcal{W}}\left(\sum_{k\in\mathcal{K}}\frac{z^{jk}\varepsilon^j T}{2}\right) + \sum_{i\in\mathcal{W}}u^i\kappa^i + \sum_{j\in\mathcal{W}}r^j\lambda^j \tag{16}$$

Subject to:

$$\sum_{i\in\mathcal{W}} u^i \le \delta_a \tag{17}$$

$$\sum_{j\in\mathcal{W}} r^j \le \delta_b \tag{18}$$

$$\sum_{s\in\mathcal{S}} x^{si} \le \sum_{k\in\mathcal{K}} D^k u^i \quad i \in \mathcal{W} \tag{19}$$

$$\sum_{i'\in\mathcal{W}\backslash\{i\}} q^{ii'} + \sum_{i'\in\mathcal{W}\backslash\{i\}} q^{i'i} \le \sum_{k\in\mathcal{K}} D^k u^i \quad i \in \mathcal{W} \tag{20}$$

$$\sum_{j\in\mathcal{W}} y^{ij} \le \sum_{k\in\mathcal{K}} D^k u^i \quad i \in \mathcal{W} \tag{21}$$

$$\sum_{k\in\mathcal{K}} z^{jk} \le \sum_{k\in\mathcal{K}} D^k r^j \quad j \in \mathcal{W} \tag{22}$$

$$\sum_{s\in\mathcal{S}} x^{si} + \sum_{i'\in\mathcal{W}\backslash\{i\}} q^{i'i} = \sum_{j\in\mathcal{W}} y^{ij} + \sum_{i'\in\mathcal{W}\backslash\{i\}} q^{ii'} \quad i \in \mathcal{W} \tag{23}$$

$$\sum_{i\in\mathcal{W}} y^{ij} = \sum_{k\in\mathcal{K}} z^{jk} \quad j \in \mathcal{W} \tag{24}$$

$$\sum_{k\in\mathcal{K}} z^{jk} \le U^j \quad j \in \mathcal{W} \tag{25}$$

$$\sum_{j\in\mathcal{W}} z^{jk} = D^k \quad k \in \mathcal{K} \tag{26}$$

$$x^{si}, y^{ij}, z^{jk}, q^{ii'} \in \mathbb{N} \quad s \in \mathcal{S}, i \in \mathcal{W}, i' \in \mathcal{W}\backslash\{i\}, j \in \mathcal{W} \tag{27}$$

$$u^i, r^j \in \{0,1\} \quad i, j \in \mathcal{W} \tag{28}$$

The objective (16) is to minimize the total distribution cost of the import product allocation network in the distribution period, which consists of the transportation cost from suppliers to TDW, transshipment cost between TDWs, distribution cost from TDW to DOW, expected delivery cost, expected inventory cost, and fixed rental cost of TDW and DOW, respectively. Constraints (17) and (18) prescribe the maximum selectable number of TDW and DOW. Constraints (19)–(22) are logical constraints, indicating that if a warehouse is not selected, the product will not be transported through that facility. Constraint (23) ensures the flow balance of transportation quantity for each TDW. Constraint (24) ensures the flow balance of delivery quantity for each DOW, which can be seen as the demand fulfillment process. Constraint (25) limits the product inventory of each TDW and DOW not to exceed its inventory capacity after distribution. Constraint (26) indicates that the forecasting demand of all consumers must be strictly satisfied in the distribution period. Constraints (27) and (28) define the properties of the variables.

5.2. Solution approach

The core of the proposed MILP model lies in the location and distribution scheme optimization for the sales period in the future. We build an iterative optimization process by embedding the MILP model into the demand forecasting model, which makes the optimal solution exploration through the commercial solvers more time-consuming. Hence, we employ the Benders Decomposition (BD) algorithm to upgrade the solution performance. The BD algorithm is an extraordinary exact solution algorithm proposed by Benders [54] and widely utilized to solve a series of MILP problems, such as location, routing, inventory management, and network design. The BD algorithm decomposes the MILP model into a master problem (MP) and a subproblem (SP) and then solves them by utilizing the solution of one in the other iteratively [55]. The proposed MILP model can be decomposed into two parts, including pure integer programming (binary variables) and linear programming (continuous variables), regarded as the MP and SP, respectively. The MP involves integer programming with two binary variables to obtain a lower bound for the overall objective and the corresponding binary values (locations of TDW and DOW), facilitating the interaction of SP. The SP represents linear programming where all binary variables are fixed to optimize the continuous variables and obtain the optimal distribution scheme, which will be solved by its dual to get an upper bound for the overall problem. If the break criterion is not satisfied, a Benders cut that includes the added continuous variable is generated using the dual SP solution. Then, the Benders cut is added to MP in the next iteration to update a new lower bound, which should be guaranteed not to be worse than the previous lower bound [41]. The stopping criterion is a small optimality gap between the optimal upper and lower bounds, and the iteration is terminated when this condition is satisfied, and the BD algorithm outputs the optimal solution.

From the objective function (16), it can be noted that the binary variables $\boldsymbol{u}$ and $\boldsymbol{r}$ are modeled to resolve the location problem of TDW and DOW, while the continuous variables $\boldsymbol{x}$, $\boldsymbol{y}$, $\boldsymbol{z}$, and $\boldsymbol{q}$ are built to address the transshipment and distribution routing problems. For known locations (fixed $\boldsymbol{u}$ and $\boldsymbol{r}$), the subsequent routing problem is linear programming that can be solved accordingly. Thus, we decompose the proposed model into an MP that involves the binary location decisions and an SP that includes the distribution routing of imported commodities under given locations of TDW and DOW.

5.2.1. Benders subproblem

To generate a reformulation of the proposed model using the BD algorithm, we first present the formulation of SP. For given binary location variables $\left(\bar{\boldsymbol{u}}, \bar{\boldsymbol{r}}\right)$ the subproblem SP $\left(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z}, \boldsymbol{q} \middle| \bar{\boldsymbol{u}}, \bar{\boldsymbol{r}}\right)$ can be regarded as a minimization problem that optimizes the distribution routing between selected warehouses. We notice that constraints (22) and (25) indicate that a warehouse is allowed to deliver commodities to consumers if it has been rented and has a delivery capacity less than the inventory capacity, respectively. These two constraints can be integrated and rewritten as:

$$\sum_{k\in\mathcal{K}} z^{jk} \le U^j r^j \quad j \in \mathcal{W} \tag{29}$$

Therefore, we reformulate the SP as:

$$\begin{aligned}(\text{SP})\ \min & \sum_{s\in\mathcal{S}}\sum_{i\in\mathcal{W}}\left(\mu^s\theta^{si}x^{si}\right) + \sum_{i\in\mathcal{W}}\sum_{i'\in\mathcal{W}\backslash\{i\}}\left(\mu^i\theta^{ii'}q^{ii'}\right) + \sum_{i\in\mathcal{W}}\sum_{j\in\mathcal{W}}\left(\eta^i\theta^{ij}y^{ij}\right) \\ & + \sum_{j\in\mathcal{W}}\sum_{k\in\mathcal{K}}\left(c^{jk} + \frac{\varepsilon^j T}{2}\right) z^{jk}\end{aligned} \tag{30}$$

subject to Eqs. (19)–(21), (29), (23)–(24), (26)–(27).

5.2.2. Dual subproblem

We define the dual variables ϕ_1^i, ϕ_2^i, ϕ_3^i, ϕ_4^j, ϕ_5^i, ϕ_6^j, ϕ_7^k associated with Eqs. (19)–(21), (29), (23)–(24), (26), and the dual of subproblem (DSP) can be given as below.

$$\begin{aligned}(\text{DSP})\ \max & \sum_{k\in\mathcal{K}} D^k\phi_7^k - \sum_{i\in\mathcal{W}}\sum_{k\in\mathcal{K}} D^k\bar{u}^i\phi_1^i - \sum_{i\in\mathcal{W}}\sum_{k\in\mathcal{K}} D^k\bar{u}^i\phi_2^i \\ & - \sum_{i\in\mathcal{W}}\sum_{k\in\mathcal{K}} D^k\bar{u}^i\phi_3^i - \sum_{j\in\mathcal{W}} U^j\bar{r}^j\phi_4^j\end{aligned} \tag{31}$$

Subject to:

$$\phi_5^i - \phi_1^i \le \mu^s\theta^{si} \quad s \in \mathcal{S}, i \in \mathcal{W} \tag{32}$$

$$\phi_5^i - \phi_5^{i'} - \phi_2^i \le \mu^i\theta^{ii'} \quad i \in \mathcal{W}, i' \in \mathcal{W}\backslash\{i\} \tag{33}$$

$$\phi_6^j - \phi_3^i - \phi_5^i \le \eta^i\theta^{ij} \quad i \in \mathcal{W}, j \in \mathcal{W} \tag{34}$$

$$\phi_7^k - \phi_4^j - \phi_6^j \le c^{jk} + \frac{\varepsilon^j T}{2} \quad j \in \mathcal{W}, k \in \mathcal{K} \tag{35}$$

$$\phi_1^i, \phi_2^i, \phi_3^i, \phi_4^j \ge 0; \phi_5^i, \phi_6^j, \phi_7^k \text{ free} \quad i \in \mathcal{W}, j \in \mathcal{W}, k \in \mathcal{K} \tag{36}$$

5.2.3. Benders cut and master problem

Generally, overseas retailers would prepare sufficient inventory and ship to the rented warehouse where the bonded zones are located at the import port in advance, as the cost of out-of-stock inventory tends to be more expensive. Although we define that consumer demand should be strictly met, all rented warehouses have enough capacities to meet all potential demand in the future sales season, i.e., out of stock is not considered. In other words, even if the distribution strategy is not implemented, the only bonded warehouse currently rented by the overseas retailer can still cover all consumers' demand, that is, the current mode of one warehouse deliver product to every consumer point across the country. Thus, we add a valid condition into MP to ensure the capacity of all rented warehouses to have sufficient inventory capacity to stock up for the forthcoming sales season, as shown in Eq. (37).

$$\sum_{j\in\mathcal{W}} r^j U^j \ge \sum_{k\in\mathcal{K}} D^k \tag{37}$$

Accordingly, for any given binary variables, i.e., $\boldsymbol{u}$ and $\boldsymbol{r}$, the Benders SP is always feasible and bounded. Hence, only one type of Benders cuts, optimality cut, will be generated and added to the MP for iteration. Let $\tilde{\phi}_1^i, \tilde{\phi}_2^i, \tilde{\phi}_3^i, \tilde{\phi}_4^j, \tilde{\phi}_5^i, \tilde{\phi}_6^j, \tilde{\phi}_7^k$ correspond to an extreme ray of the polyhedron represented by Eqs. (32)–(36) and used as the

optimality cut:

$$\sum_{k\in\mathcal{K}} D^k \tilde{\phi}_7^k - \sum_{i\in\mathcal{W}}\sum_{k\in\mathcal{K}} D^k u^i \tilde{\phi}_1^i - \sum_{i\in\mathcal{W}}\sum_{k\in\mathcal{K}} D^k u^i \tilde{\phi}_2^i - \sum_{i\in\mathcal{W}}\sum_{k\in\mathcal{K}} D^k u^i \tilde{\phi}_3^i - \sum_{j\in\mathcal{W}} U^j r^j \tilde{\phi}_4^j \le \xi \tag{38}$$

Utilizing the above representations, the Benders MP can be reformulated as follows:

$$\text{(MP)}\ \min\ \xi + \sum_{i\in\mathcal{W}} u^i \kappa^i + \sum_{j\in\mathcal{W}} r^j \lambda^j \tag{39}$$

subject to (17)–(18), (28), (37), and (38). Santoso et al. [56] proved that once a good upper bound for BD algorithm is generated, integrating knapsack inequalities with optimality cut (38) can achieve considerable improvement on solution quality. Let UB_p and LB_p represent the best known upper- and lower-bound found until iteration p, and the following knapsack inequalities are added to MP:

$$\sum_{i\in\mathcal{W}} u^i \kappa^i + \sum_{j\in\mathcal{W}} r^j \lambda^j + \sum_{k\in\mathcal{K}} D^k \tilde{\phi}_7^k - \sum_{i\in\mathcal{W}}\sum_{k\in\mathcal{K}} D^k u^i \tilde{\phi}_1^i - \sum_{i\in\mathcal{W}}\sum_{k\in\mathcal{K}} D^k u^i \tilde{\phi}_2^i - \sum_{i\in\mathcal{W}}\sum_{k\in\mathcal{K}} D^k u^i \tilde{\phi}_3^i - \sum_{j\in\mathcal{W}} U^j r^j \tilde{\phi}_4^j \le UB_p \tag{40}$$

$$LB_p \le \xi + \sum_{i\in\mathcal{W}} u^i \kappa^i + \sum_{j\in\mathcal{W}} r^j \lambda^j \tag{41}$$

The pseudocode of the BD algorithm for solving the proposed model is summarized in Algorithm 2.

Algorithm 2: BD algorithm

Data: supplier set $\mathcal{S}$, candidate warehouse set $\mathcal{W}$, consumer set $\mathcal{K}$, maximum selectable number δ_a and δ_b, and other parameters.
Result: location and distribution plan $(\boldsymbol{u}', \boldsymbol{r}', \boldsymbol{x}', \boldsymbol{y}', \boldsymbol{z}', \boldsymbol{q}')$, and objective value π'.

Initialize: set $UB = +\infty$, $LB = 0$, $\epsilon = 0.01$, $p = 0$, $p_{\max}$, and $BENDERS\ CUTSET$ empty;
while $p < p_{\max}$ **do**
  Solve MP and obtain objective value π^{MP} and location plan (u, r);
  $LB \leftarrow \max\{LB, \pi^{MP}\}$;
  Solve DSP and obtain objective value π^{DSP} and distribution plan (x, y, z, q);
  $UB \leftarrow \min\{UB, \pi^{DSP} + \pi^{MP} - \Theta(CUTSET)\}$;
  if $\left|\frac{UB-LB}{UB}\right| \le \epsilon$ **then**
    break *Output UB as objective value and the corresponding solution set*;
    else
      Add the optimality cut to BENDERS CUTSET
    end
  end
  $p = p + 1$;
end

5.2.4. FDIO process

The expected delivery time of some consumers who are closer to several leased bonded warehouses will be significantly reduced after completing the optimization procedure for the location and allocation of TDWs and DOWs. The improvement of logistics timeliness may further stimulate the increase in potential demand of online consumers. Thus, the expected delivery time should be modified and re-input the CNN-LSTM model again to generate the newly updated demand forecasting result. This iterative optimization procedure is carried out round after round until the entire distribution network no longer changes, and then, the corresponding warehouse rental plan and product allocation scheme are outputted as the optimal results. The brief description of the interrelated interdependence for the FDIO process is shown as Algorithm 3.

Algorithm 3: FDIO process

Data: supplier set $\mathcal{S}$, candidate warehouse set $\mathcal{W}$, consumer set $\mathcal{K}$, maximum selectable number δ_a and δ_b, and other parameters.
Result: forecasting demand D^{k*}, optimal location plan $(\boldsymbol{u}^*, \boldsymbol{r}^*)$, distribution plan $(\boldsymbol{x}^*, \boldsymbol{y}^*, \boldsymbol{z}^*, \boldsymbol{q}^*)$, and total cost π^*.

Initialize: set consumer demand D_0^k (initialization forecasting result of prediction model);
while $g = 1$ *and* $g < g_{\max}$ **do**
  Optimize $(u, r, x, y, z, q) \leftarrow D_{g-1}^k$;
  Update $D_g^k \leftarrow$ CNN-LSTM-GA Algorithm;
  if $D_{g-1}^k = D_g^k$ *or* $(u, r)_g = (u, r)_{g+1}$ **then**
    break *Output the objective value and the corresponding solution set*;
    else
      set $g = g + 1$ and go to step 2;
    end
  end
end

6. Case study

6.1. Data description and preparation

This section investigates a leading online personal care product retailer as the case to verify the proposed model. Overseas retailers usually choose centralized freight for cost considerations in long-distance ocean shipping, which means that one batch of products is shipped to one port rather than multi-batch and multi-destination ports. In the case study, this retailer already adopts the bonded warehouse logistics mode for import business and transports the products to Hangzhou Port in China, which is regarded as the initial point, so there is only one supplier. More than twenty thousand transaction data and two thousand efficient comments for this retailer's best-selling shower gel product over the past two years are collected. Since the collected data that measure the purchasing power of consumers are the city-level dimension, we divide the sales destinations into city-level (including 293 prefecture-level municipalities and four municipalities directly under the central government) to maintain consistency in data dimensions. Table 2 shows the summary statistics of the collected and collated input data for training the prediction model.

In terms of the data for input parameters, we collect the latitude and longitude of cities with bonded warehouses and calculate the distance between each warehouse node through Euclidean distance. The transportation cost is estimated based on some full truck load servicers such as YunManMan and Lalamove in China, assuming the value 0.001 for any μ and η. The delivery cost between each warehouse node and each consumer point is quantified as a linear combination of $c^{ik} = a + \max\{b\theta^{ik}, c_0\}$, where $a = 2.5$ denotes the fixed package cost, $b = 0.0075$ represents the unit distance transport cost, and $c_0 = 6$ is the fixed minimum cost. The monthly rental cost for each candidate warehouse is randomly generated in the range of 1500 to 3000 yuan with 100 yuan interval. Similarly, the daily inventory cost for unit product is randomly generated in the range of 0.02 to 0.05 yuan with 0.005 yuan interval. The storage capacity of each bonded warehouse is assumed to depend on the political level and economic development level of the city where the warehouse is located, which is randomly generated within four ranges: [500,1000], [1000,2000], [2000,4000], and [4000,8000]. Additionally, the fixed cost for transshipping inventory through a warehouse is assumed to be the half of its monthly rental cost.

Table 2
Summary statistics of the preprocessed data.

Category	Description	2021/01	2022/12
Comment	Modified score	4.7754	4.9323
Product	Price	44.9	49.9
	Sales volume	Beijing-290	Beijing-225
		Tianjin-133	Tianjin-169
		...	...
		Kashgar-9	Kashgar-12
Operational status	Number of fans (10,000)	42.3	51.6
	Damage rate (%)	1.749	1.856
	Promotional activity (0–6)	0	0
City-level indicator	Population (10,000)	Beijing-2,189.3	Beijing-2,184.3
		Tianjin-1,387.0	Tianjin-1,363.0
		...	...
		Kashgar-70.69	Kashgar-71.13
	Disposal income (yuan)	Beijing-69,433.5	Beijing-77,415.0
		Tianjin-43,854.1	Tianjin-48,976.0
		...	...
		Kashgar-16,392.0	Kashgar-17,245.0
	Urbanization (%)	Beijing-87.55	Beijing-87.60
		Tianjin-84.70	Tianjin-85.11
		...	...
		Kashgar-46.80	Kashgar-47.13
Logistics efficiency	Delivery time (hour)	Beijing-42.3	Beijing-43.1
		Tianjin-40.7	Tianjin-39.6
		...	...
		Kashgar-82.3	Kashgar-81.6

6.2. Forecasting results

To estimate the performance of forecasting results, we compare the proposed model with other classic prediction models, such as CNN, LSTM, autoregressive integrated moving average model (ARIMA), back propagation neural network (BP), and the basic version CNN-LSTM optimized hyperparameters. Additionally, we also compare the performance and stability of proposed forecasting model without integrating the modified comment score data, i.e., CNN-LSTM-GA* in Tables 3 and 4, to verify the effectiveness of the collated dataset. We adopt two commonly used evaluation functions for performance assessment, along with the comparison with other approaches, namely, mean absolute percentage error (MAPE) and root mean square error (RMSE), which are shown as below. The smaller the MAE or RMSE values, the better demand prediction results.

$$\begin{cases} \text{MAPE} = \frac{1}{K'} \sum_{k' \in K'} \left| \frac{D^{k'} - \sum_{t \in T} d_t^{k'}}{\sum_{t \in T} d_t^{k'}} \right| \times 100\% \\ \text{RMSE} = \sqrt{\frac{1}{K'} \sum_{k' \in K'} \left(D^{k'} - \sum_{t \in T} d_t^{k'} \right)^2} \end{cases} \tag{42}$$

where $K' = 31$ represents the total thirty-one provinces and autonomous regions in China, $D^{k'}$ denotes the aggregated actual consumers' demands in region k', and $d_t^{k'}$ is the aggregated forecasting consumers' demands for period t in region k'. The preprocessed time series data set is divided into a training set and two testing sets. The training set contains 18 months of time series data from January 2021 to June 2022, and the testing set contains six months of time series data from July 2022 to December 2022. Although the output of the prediction model is the forecasting demand in the dimension of month, calculating the performance of the prediction model with a shorter time dimension (i.e., days or weeks) will result in a larger deviation value, even though the time series dimension of the test set can be selected freely. Generally, an overseas retailer's distribution plan is to prepare as much inventory as possible for sales period in the future (such as quarterly or even longer). Thus, the prediction model can be regarded as the more effective as long as the total deviation within the total sales period is smaller. Additionally, it is extremely difficult to accurately predict the future demand of consumers for express delivery services in nearly three hundred cities as long as the delivery within the province can basically meet the logistics fulfillment of the next-day delivery. Thus, we measure the performance of the forecast models to focus on the provincial dimension, that is, $K' = 31$ provinces and autonomous regions (excluding Hong Kong, Macao, and Taiwan).

For comparison, we run each forecasting model for twenty rounds and average the performance results on two indicators of different prediction methods in various time dimensions for the testing set, shown in Table 3. The MAPE and RMSE values of the proposed deep learning approach in the semiannual dimension of the prediction result are 32.8 and 44.0, which are the lowest compared to other methods, but the accuracy improvement by integrating the comment score modification process seems to be not significant. Moreover, the forecasting accuracy and stability of future demand in the monthly and quarterly dimensions are also better than the other five forecasting methods. We also observe that the prediction accuracy of the seven methods is generally lower in November and December than that of the other four months because the e-commerce platform has launched the two annual large-scale promotions of Double Eleven (November 11th) and Double Twelve (December 12th) in November and December, respectively. Although the promotions have caused a significant impact on prediction performance, the proposed CNN-LSTM-GA forecasting model still outperforms other methods in terms of prediction accuracy over these two months. Additionally, we also display the predictive stability of different forecasting models, which is expressed as the variance of each prediction model for total twenty rounds on two assessment indicators, shown in Table 4. The results show that although the inclusion of the comment score modification process brings limited marginal improvement to the forecasting accuracy of CNN-LSTM-GA, but it can improve the predictive stability more prominently.

6.3. Distribution results

In this subsection, we construct different test sets from small- to large-scale cases to verify the effectiveness of the proposed model and algorithm. Specifically, we start by constructing the smallest-scale case, i.e., only the provincial capital cities and municipalities directly under the central government with comprehensive bonded zone and rentable bonded warehouses are selected as the set of warehouse candidate locations. In the largest-scale case, we select all cities with comprehensive bonded zone and rentable bonded warehouses that can be selected as the set of warehouse candidate locations. All these instances are tested based on the real-life demand distribution from July 2022 to September 2022 in the previous section. Notably, all test cases in this section are static, i.e., demand distribution results are given and fixed, and the iterative test results integrating constantly updated demand and distribution network are shown in the subsequent section.

First, we constructed a smallest-scale case and applied it to the case study to verify the effectiveness of the algorithm. The smallest-scale case consisted of 30 candidate warehouse locations with comprehensive bonded zones and rentable bonded warehouses, including 22 provincial capitals, four municipalities directly under the central government, and four autonomous regional capitals (except Tibet, which had no rentable bonded warehouse). When the distribution strategy is not implemented, all consumer demand is satisfied by the bonded warehouse in the comprehensive bonded zone, with a total operation cost of 179,953 yuan. Table 5 lists the comparative results for changing the maximum selectable number of TDW and DOW, where the fourth and eighth column refers to the results of the objective function. As shown, the horizontal transshipment of inventory between rented TDWs and distribution to DOWs reduces the average delivery cost of each commodity, leading to a more significant reduction in the total operation cost. The solution results of the BD algorithm and CPLEX solver reveal that consistent optimal solutions can be obtained in small-scale cases. Additionally, an interesting result is that the optimal objective value 169,080 never changes when δ_a and δ_b are greater than 4 and 10 simultaneously. Because the optimal warehouse locations and distribution plan have been achieved given a set of candidate warehouse locations $W = 30$,

Table 3
Performance assessment of forecasting models.

Period	CNN		LSTM		ARIMA		BP		CNN-LSTM		CNN-LSTM-GA*		CNN-LSTM-GA	
	MAPE	RMSE	MAPE	RMSE	MAPE	RMSE	MAPE	RMSE	MAPE	RMSE	MAPE	RMSE	MAPE	RMSE
07	30.4	23.3	34.5	25.9	36.2	29.2	33.7	24.0	25.2	18.3	23.2	17.9	24.1	18.0
08	32.7	26.4	33.5	28.4	31.1	25.2	39.8	28.7	26.5	20.2	25.9	18.4	23.0	17.8
09	35.4	27.5	39.1	30.1	35.0	26.6	45.6	34.4	26.1	19.2	24.0	17.7	21.5	16.3
10	29.9	23.7	32.7	24.9	35.6	26.9	38.2	28.5	28.7	19.0	22.8	17.3	23.6	17.6
11	40.7	30.1	39.6	27.0	46.1	34.6	43.3	31.6	31.8	22.6	29.0	19.0	26.3	18.2
12	37.8	29.4	44.2	30.4	42.4	30.2	45.3	32.7	31.9	22.3	30.4	19.6	25.9	19.3
07–09	34.1	35.5	28.2	27.8	46.0	43.3	36.7	38.7	29.6	31.1	26.8	27.1	22.8	23.0
10–12	43.9	44.3	42.0	43.4	45.2	48.4	45.5	49.6	33.9	31.2	24.7	26.4	25.0	26.5
07–12	40.9	51.5	44.1	56.6	49.4	60.1	50.7	62.4	38.3	48.0	33.3	44.6	32.8	44.0

Table 4
Stability assessment of forecasting models.

Period	CNN		LSTM		ARIMA		BP		CNN-LSTM		CNN-LSTM-GA*		CNN-LSTM-GA	
	MAPE	RMSE	MAPE	RMSE	MAPE	RMSE	MAPE	RMSE	MAPE	RMSE	MAPE	RMSE	MAPE	RMSE
07	7.2	4.9	7.3	5.0	7.6	5.5	7.5	5.4	6.7	4.7	6.3	4.3	5.7	4.2
08	4.8	3.8	5.1	4.1	5.7	4.5	5.4	4.4	54	3.5	5.5	3.3	4.8	3.4
09	7.0	4.7	7.3	4.6	7.2	5.0	7.1	5.1	6.6	4.5	5.9	3.9	5.4	3.8
10	5.0	5.4	5.4	5.3	6.9	5.8	7.0	5.8	5.6	4.6	5.1	4.6	5.0	4.0
11	5.9	4.1	5.5	4.4	5.9	5.2	6.3	5.5	4.9	4.2	4.5	4.1	4.5	3.9
12	6.5	6.0	6.5	6.5	6.9	5.6	7.2	5.9	7.1	5.5	6.6	5.6	5.9	4.7
07–09	7.9	7.3	7.8	7.0	7.9	7.4	8.5	7.5	7.3	6.9	7.1	6.6	6.1	5.1
10–12	8.4	7.8	8.2	7.4	8.6	8.3	8.7	7.8	7.4	7.1	7.1	6.7	6.3	5.6
07–12	9.2	8.4	9.0	8.3	9.7	8.8	10.1	9.1	8.3	7.6	7.8	7.0	7.1	6.2

Table 5
Comparison of smallest-scale test cases for distribution optimization.

$\mathcal{W}$	δ_a	δ_b	Obj. (CNY)	$\mathcal{W}$	δ_a	δ_b	Obj. (CNY)
30	2	10	175,025	30	4	10	169,080
30	2	15	174,738	30	4	15	169,080
30	2	20	174,553	30	4	20	169,080
30	2	25	174,553	30	4	25	169,080
30	3	10	173,517	30	5	10	169,080
30	3	15	171,232	30	5	15	169,080
30	3	20	171,232	30	5	20	169,080
30	3	25	171,232	30	5	25	169,080

the total operation cost is reduced by approximately 6.04% compared to the no-distribution strategy. Further, we present the optimal location and distribution plan, which involves four TDWs and eight DOWs under the given candidate set, as shown in Fig. 3.

Then, we test the performance of the BD algorithm in different scale cases by increasing candidate warehouse locations from 30 to 100 and setting the maximum selectable numbers of TDW and DOW to the total number of candidate warehouses (i.e., $\delta_a = \delta_b = W$). Additionally, we set a discount factor α of the transshipment cost between warehouses to reflect the economics of scale, which is a common assumption in relevant studies [57,58]. Under the given objective convergence gap $\epsilon = 0.01$, the detailed results of which are shown in Table 6, where the last two column represents the computational time of Benders decomposition algorithm and solver, respectively. The BD algorithm can find the global optimal solution within five minutes for all cases with $W \leq 100$, while the solver mostly consumes more than twice the time to find the corresponding optimal solution. In larger scale test cases of candidate set, both BD algorithm and solver can output the optimal solution, revealing the good solution performance in terms of both effectiveness and stability for solving the proposed model. However, the computational time of the BD algorithm is much lower than that of solver on different scale of candidate warehouses, which may due to the fact that the knapsack inequalities with optimality cut result in an improvement on the solution quality as well as the valid inequality guarantees the feasibility of the solution in each iteration to ensure the computational efficiency of the BD algorithm on solving different scale instances. Overall, the BD algorithm applied in optimizing the proposed model has significant adaptability and efficiency. Although

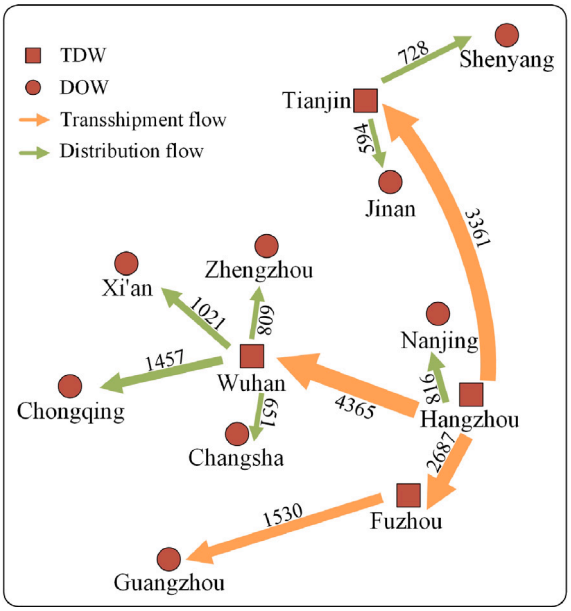


Fig. 3. Optimal location and distribution plan of smallest-scale case.

using solver to optimize the largest-scale case in the static model is not particularly time-consuming, the solution efficiency would not be

Table 6
Comparison of different candidate sets for distribution optimization.

α	W	Obj. (CNY)	BD (s)	CPLEX (s)	α	W	Obj. (CNY)	BD (s)	CPLEX (s)
0.2	30	161,475	4.9	13.2	0.5	30	165,354	8.2	17.9
	40	160,993	9.3	25.6		40	164,818	29.1	76.8
	50	160,355	14.5	30.8		50	164,542	3.1	12.0
	60	159,147	14.8	57.3		60	163,043	19.0	60.3
	70	157,912	29.1	68.8		70	161,957	32.4	102.7
	80	157,583	57.1	173.5		80	161,593	105.8	298.9
	90	157,701	164.4	379.1		90	161,742	227.9	586.4
	100	157,614	246.7	519.8		100	161,607	265.0	742.1
0.7	30	167,753	6.4	23.8	1.0	30	169,080	5.3	19.7
	40	167,191	9.0	32.5		40	168,479	7.5	29.8
	50	166,646	5.4	14.0		50	167,354	13.0	28.2
	60	165,068	68.0	146.9		60	166,232	16.4	38.7
	70	164,209	47.6	163.1		70	165,766	40.7	148.0
	80	163,951	135.1	480.8		80	165,434	104.5	225.7
	90	164,029	245.6	723.7		90	165,288	76.7	290.3
	100	163,892	284.9	866.1		100	165,150	107.2	345.9

further guaranteed if the static distribution model is embedded with the prediction model to make up an iterative optimization process. Hence, the next subsection represents the significant improvement in the solution efficiency of the BD algorithm in the iterative optimization process.

6.4. FDIO results

When the overseas retailer develops a warehouse rental plan and corresponding inventory distribution scheme based on the prediction demand, the logistics timeliness of some consumers will be significantly improved, which may affect the redistribution of potential demand and thus lead to the formulated plans may fail to achieve the optimal. In this subsection, we integrate the forecasting-distribution optimization process to solve the optimal strategy iteratively and further verify the effectiveness based on the case study in the above sections.

The length of the expected sales period will significantly affect the warehouse rental plan and commodity distribution scheme. We assume that the retailer develops four fulfillment plans to meet the sales period of different durations from July to December in 2022: P1 (July to September), P2 (July to October), P3 (July to November), P4 (July to December). Under the embedded iterative optimization procedure, we increase the maximum running time of the BD algorithm and solver to four times of the previous static cases and first test the performance of the proposed algorithms under different cases in Table 7. After embedding the distribution model into the prediction model to form an iterative optimization process, the solver is unable to obtain the optimal solution within one hour when there are 100 candidate warehouses, while the BD algorithm is more efficient in outputting the optimal solution when the number of candidate warehouses is less than 90 within 10 min. Compared to the static optimization procedure in the previous section, the solution efficiency of the BD algorithm is further amplified during the FDIO process, which effectively supports the design of an efficient distribution network.

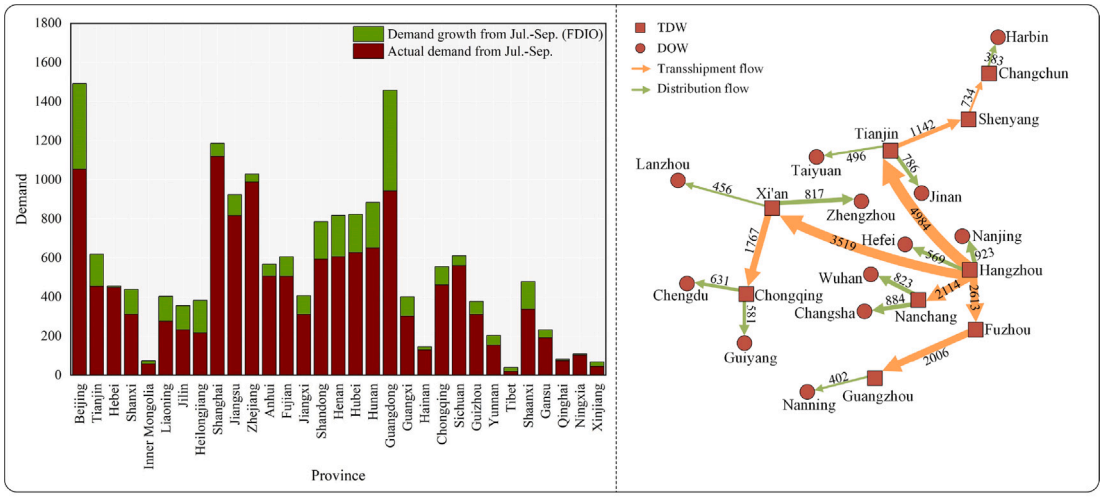
Then, we present the provincial actual demand for period P1 and period P4 and the corresponding demand growth volume when FDIO process is implemented, shown in Figs. 4(a) and 5(a), respectively. As seen, the implementation of the FDIO significantly promote the demand growth in both Guangdong Province and Beijing. Because they are economically developed regions, and the distribution of inventory to nearby rented warehouses greatly improve logistics efficiencies, thereby stimulating potential demand. Further, in order to show the comparative results of distribution plan, we visualize the optimal warehouse rental plan and distribution scheme of these two periods under smallest-scale and largest-scale candidate set, shown in Figs. 4(b) and 5(b). It can be seen from Fig. 4(b) that the optimal solution for scenario P1 is to rent nine warehouses as TDWs and twelve warehouses as DOWs. Since the retailer needs to satisfy the future demand from July to December in scenario P4, the increase in consumer demand results in renting more warehouses. Fig. 5(b) shows the optimal solution under the largest-scale candidate set, that is, fourteen warehouses are rented as TDWs and nineteen warehouses are rented as DOWs. In addition, the warehouse rental plan in some provinces significantly varies across different candidate set and planning periods. For instance, as seen in Figs. 4(b) and 5(b), the DOW in Shandong Province has been changed from Jinan warehouse to Qingdao warehouse. Guiyang warehouse has been transformed from DOW to TDW. One more bonded warehouse has been rented as a DOW in Fujian Province, Guangdong Province, etc. The comparative results further verify the effectiveness of the proposed model.

Table 8 presents the detailed operation performance of the benchmark (no-distribution strategy) and FDIO process, which summarizes the demand, total operation cost, average cost for each product and overall profit, and res the increase in demand and profit for FDIO compared to the benchmark case. Specifically, the increase in logistics efficiency brought about by distribution may lead to an increase of approximately 25.4% in total demand from July to September. According to the detailed data provided by the retailer, the average unit cost of this shower gel product is about 50% of the selling price (including the wholesale and overseas headway transportation costs). While keeping the selling price of goods unchanged at 49.9 yuan and the per unit cost 50% of the selling price, the total profit can be increased by approximately 12.6% compared to the actual benchmark case with the no-distribution strategy. Surprisingly, although renting warehouses and inventory distribution under the FDIO process may incur additional warehouse rental costs, distribution transportation costs, etc., the distribution process significantly increases the potential demand, resulting in approximately 25% higher than the no-distribution scenario. Although more decentralized warehouses can improve the logistics service fulfillment efficiency to boost total demand, it may not effectively increase the profit as the inventory cycle becomes longer, leading to higher setup, transshipment, and distribution costs. As Table 8 shows, the per unit product fulfillment cost is 1.03–1.85 yuan higher than the benchmark scenario. Interestingly, the retailer in the P2 scenario has increased profit margin and reduced unit product cost compared to the P1 scenario because the sales margin improvement from redeployment is higher than the cost burden caused by increasing distribution volume to satisfy the future demand in October. To meet the longer sales period, which results in a greater inventory allocation, the retailer may be forced to rent more warehouses and increase the number of trunk transportation routes, making the positive effect of the sales boost weaker than the negative effect of the cost increase. For instance, as the inventory cycle lengthens from 4 to 6 months, the profit improvement ratio of FDIO compared to the benchmark scenario decreases from 12.9% to 5.2%, calculated based on the constant inventory cost. It is worth noting that the promotion activities in November and December

Table 7
Comparison of different candidate sets for FDIO process under four periods.

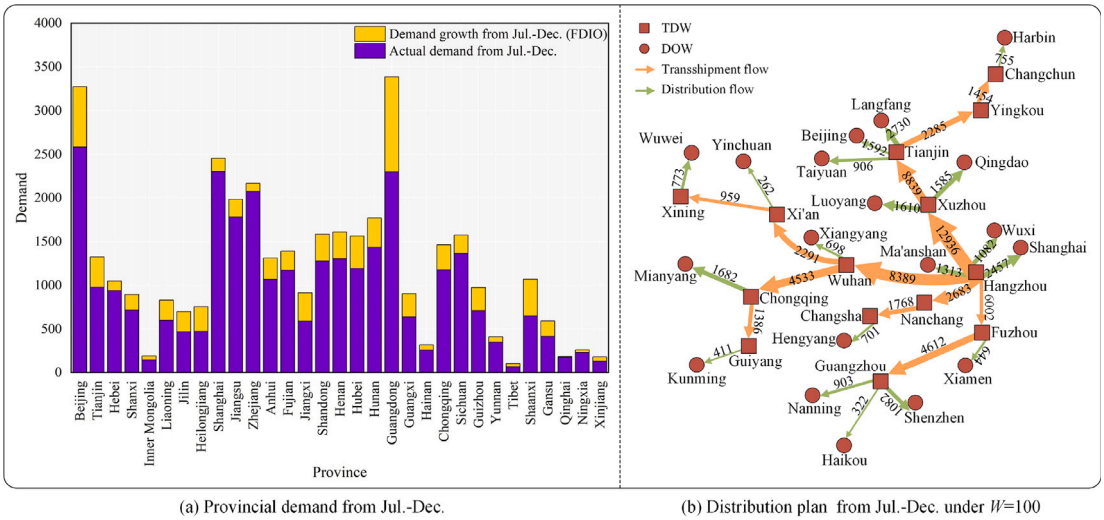
Period	W	Obj. (CNY)	BD (s)	CPLEX (s)	Period	W	Obj. (CNY)	BD (s)	CPLEX (s)
P1	30	253,602	19.8	50.7	P2	30	304,157	21.0	51.2
	40	251,227	29.8	67.8		40	302,969	34.4	78.3
	50	248,993	48.5	108.3		50	300,546	51.5	100.7
	60	246,762	98.4	150.4		60	299,409	103.1	157.5
	70	246,110	155.5	292.4		70	297,023	154.4	290.4
	80	245,998	264.7	649.3		80	296,610	247.5	607.1
	90	245,732	545.0	1,405.5		90	296,388	578.6	1,502.4
	100	245,732	time (2.03%)	time (3.85%)		100	296,388	time (1.74%)	time (3.98%)
P3	30	456,245	21.1	60.0	P4	30	585,718	20.8	53.7
	40	454,204	33.4	80.5		40	582,087	37.5	85.3
	50	451,487	61.6	120.7		50	580,473	52.3	100.7
	60	448,906	125.7	192.1		60	577,220	123.8	189.2
	70	447,590	192.3	361.7		70	575,015	180.4	339.3
	80	447,128	313.0	767.8		80	574,430	297.6	730.0
	90	446,971	567.2	1,545.3		90	574,294	530.9	1,475.4
	100	446,670	time (1.66%)	time (4.90%)		100	573,832	time (1.82%)	time (4.12%)

Note: The *time* in the table represents the time limit exceeded (600 s for the BD algorithm and 3,600 s for CPLEX), the percentage value in parentheses after *time* is the relative gap.



(a) Provincial demand from Jul.-Sep. (b) Distribution plan from Jul.-Sep. under W=30

Fig. 4. Provincial demand and distribution plan from Jul.-Sep.



(a) Provincial demand from Jul.-Dec. (b) Distribution plan from Jul.-Dec. under W=100

Fig. 5. Provincial demand and distribution plan from Jul.-Dec.

may affect warehouses to increase the inventory cost during the campaigns, so the profit improvement rates are likely to be further decreased. However, to satisfy the long-term sales target, the retailer needs to consider that multi-frequency transshipment and allocation are necessary if a relatively short distribution and inventory cycle plan is adopted. As a result, they need to make a trade-off between the

Table 8
Expected operation performance of different period.

Period	Benchmark (no distribution)				FDIO process				Comparison	
	Demand (unit)	Obj. (CNY)	Avg. cost (CNY)	Profit (CNY)	Demand (unit)	Obj. (CNY)	Avg. cost (CNY)	Profit (CNY)	Demand increase (%)	Profit increase (%)
P1	13,510	179,953	13.32	157,801	16,938	245,732	14.51	177,721	25.4	12.6
P2	16,649	222,103	13.34	194,132	20,625	296,388	14.37	219,229	23.9	12.9
P3	23,654	320,275	13.54	271,074	29,765	446,670	15.01	297,452	25.8	9.7
P4	29,626	402,021	13.57	338,622	37,208	573,832	15.42	356,346	25.6	5.2

transshipment, distribution, and inventory costs and the expected sales revenue, as frequent trunk line transfers result in higher transportation costs.

7. Concluding remarks

This paper constructs an iterative optimization process for CBEC overseas retailers to design their characteristic distribution network for imported commodities. The valuable contribution of this work is twofold. As shown throughout the paper, we build a hybrid deep learning model to predict future demand for CBEC retailer and construct an iterative optimization process to iteratively improve the distribution network considering changes in logistics timeliness. However, it is worth noting that the optimization results of the FDIO process in our research are based on the predicted demand, and the actual impact of the distribution network formulation on the real demand improvement is difficult to measure, which needs to be observed by retailers in actual operations. However, it does not diminish the value of this work.

Specifically, we take full advantage of the available data, including comment, product, operation, logistics service, and city-level purchasing power, as the input data variables to output the transaction volume as the dependent variable. Since product reviews on e-commerce platforms are integers ranging from 1 to 5, making it difficult to reflect the impact of reviews on consumers effectively, we correct the initial integer score through the sentiment analysis to better integrate the impact utility of online reviews. Then, we propose an integrated CNN-LSTM-GA deep learning model to train the historical data and output the forecasting demand, in which the CNN and LSTM are used to extract recessive features and output time-sequential forecasting demand, and GA is adopted to optimize the hyper-parameters of CNN and LSTM. Subsequently, we build an optimization model for the distribution network design and obtain the warehouse renting and inventory distribution plan using the Benders decomposition algorithm. Since the implementation of the distribution strategy significantly improves the logistics timeliness in some consumers' locations, which may affect the redistribution of forecasting demand, we further model a forecasting-distribution iterative optimization process to seek the optimal plan. Our research presents a bridge between the deep learning model and optimization program to formulate the iterative optimization process, providing new insights into iterative optimization of CBEC import supply chain distribution networks.

The managerial insights of CBEC retailers are concluded as follows:

- The CBEC platform proposes Double Eleven and Double Twelve promotional activities in November and December, so the warehouse renting plan and inventory distribution scheme need to be prepared in advance. The warehouse rental costs and product delivery costs in these two months would generally increase, making the actual profit growth rate lower than the expected value and requiring the retailer to strategically determine the distribution volume based on actual situations.
- Our research addresses two important concerns of overseas retailers: How does the improved logistics timeliness caused by the distribution network affect the potential demand for CBEC imported commodities? How is the trade-off between the increasing demand and the total distribution cost measured? Moreover, in the actual scenario, the overseas retailers can strategically control the maximum selectable number of warehouses and dynamically adjust the prediction demand of some consumer locations to be fulfilled. For instance, the retailer can start by renting one warehouse, observing the improvement in sales volume and then deciding whether to rent another warehouse and further location and distribution volume in the next sales season.
- Integrating comment score modification process to the forecasting model may bring little marginal effect on demand predictive accuracy, but it achieves a higher improvement in the forecasting stability. Thus, the overseas retailers can integrate other factors, which usually must be manually recorded or are difficult to observe, into the deep learning model to better predict future demand. These factors, such as coupons issued by the platform, will affect the final order price, the complementarity between similar products, and the competition with other retailers.

The proposed FDIO process still presents several limitations. For example, the dimensions of the input data used for prediction could be wider, the transportation costs could be more explicit, and more efficient algorithms could be introduced to design more complex distribution networks. Researchers can extend this model to a scenario of designing multiproduct distribution network, integrating three-dimensional container loading problems and considering the joint operation of overseas direct mail and bonded warehouse modes in the future.

CRediT authorship contribution statement

Xuhui Chen: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Yong He:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization. **Golnaz Hooshmand Pakdel:** Writing – review & editing. **Chung-Hsing Yeh:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available upon reasonable request.

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